

BERGEN FLORIDA, Norway

WMO: 013170

Lat: 60.383N

Lon: 5.333E

Elev: 118

StdP: 14.63

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	18.9	22.3	5.3	7.3	28.5	10.0	9.2	30.0	29.2	39.8	25.7	40.2	3.1	140	0.604

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	77.3	62.0	73.3	60.6	70.0	59.1	64.4	72.8	62.7	70.0	61.1	67.4	8.0	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.4	81.6	67.3	59.7	76.8	65.6	58.1	72.3	64.0	29.5	72.7	28.3	70.0	27.2	67.5	72.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.3	20.2	17.7	17.0	82.7	4.0	4.0	14.1	85.6	11.7	87.9	9.5	90.1	6.6	93.0		
(5)			15.7	66.8	4.0	2.3	12.8	68.4	10.5	69.7	8.2	71.0	5.3	72.7	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.2	36.1	36.1	38.8	45.1	51.1	56.1	60.5	60.3	55.2	47.8	41.3	37.3	(6)	(7)
	DBStd	10.58	6.21	5.61	4.88	5.40	5.95	5.11	5.27	4.90	4.83	5.47	6.32	6.97	(7)	(8)
	HDD50	2170	431	390	346	167	54	5	0	0	10	105	267	395	(8)	(9)
	HDD65	6542	896	810	811	598	432	271	160	164	295	534	712	859	(9)	(10)
	CDD50	1149	0	0	0	19	89	190	327	318	166	36	4	0	(10)	(11)
	CDD65	50	0	0	0	0	2	6	22	18	2	0	0	0	(11)	(12)
	CDH74	286	0	0	0	1	22	40	142	76	5	0	0	0	(12)	(13)
	CDH80	45	0	0	0	0	3	2	32	8	0	0	0	0	(13)	(14)
(14) Wind	WSAvg	7.7	9.0	8.4	8.3	7.8	7.8	7.3	6.6	6.8	7.1	7.6	7.8	8.4	(14)	(15)
(15) Precipitation	PrecAvg	89.5	8.3	6.2	6.1	4.8	4.1	5.2	5.7	7.4	10.6	10.6	10.7	9.9	(15)	(16)
	PrecMax	122.2	19.6	13.2	16.3	8.6	9.5	10.6	11.7	14.5	24.8	21.9	22.0	22.4	(16)	(17)
	PrecMin	55.9	1.5	1.1	0.6	0.3	0.7	0.2	0.7	0.2	2.6	1.7	3.0	1.0	(17)	(18)
	PrecStd	19.0	4.7	3.3	3.6	1.9	2.0	2.6	2.7	3.3	4.9	4.9	4.5	5.6	(18)	(19)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.2	49.7	54.5	67.5	76.5	79.1	84.5	80.9	73.1	65.5	57.2	52.2	(19)	(20)
		MCWB	43.1	44.2	45.7	52.0	58.5	59.9	65.5	63.6	60.5	54.8	49.6	47.1	(20)	(21)
	2%	DB	47.3	47.0	49.8	61.8	70.5	73.2	77.9	75.7	68.4	60.4	53.7	49.6	(21)	(22)
		MCWB	43.2	43.5	42.8	49.2	55.3	58.4	62.9	62.7	57.0	52.1	48.5	46.1	(22)	(23)
	5%	DB	45.7	45.4	47.1	57.8	66.1	68.6	73.2	71.8	65.5	57.4	51.5	48.0	(23)	(24)
		MCWB	42.7	42.5	42.2	47.3	53.6	56.5	62.1	60.9	56.5	51.3	47.4	45.2	(24)	(25)
	10%	DB	44.2	43.9	45.5	54.0	61.8	65.1	69.4	68.5	62.8	55.3	49.6	46.4	(25)	(26)
		MCWB	41.6	41.4	41.4	45.7	51.3	55.3	60.1	59.6	55.5	50.4	45.9	43.7	(26)	(27)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.5	47.1	47.2	54.4	60.2	62.5	67.3	66.5	62.5	57.1	52.5	49.1	(27)	(28)
		MCDB	47.5	48.0	51.3	63.4	72.7	73.3	79.7	75.4	69.0	61.8	54.6	50.5	(28)	(29)
	2%	WB	44.9	44.8	45.4	50.6	56.7	60.0	64.7	64.6	60.2	54.6	50.4	47.4	(29)	(30)
		MCDB	46.0	46.1	48.0	58.6	67.5	69.1	74.2	71.9	64.7	58.0	52.6	48.9	(30)	(31)
	5%	WB	43.5	43.4	43.8	48.6	54.3	58.2	62.9	62.9	58.3	53.1	48.5	45.7	(31)	(32)
		MCDB	44.9	45.0	46.2	55.3	64.0	66.2	70.5	69.6	62.6	56.2	50.6	47.5	(32)	(33)
	10%	WB	41.9	41.7	42.2	47.0	52.3	56.5	61.3	61.1	56.6	51.3	46.6	44.0	(33)	(34)
		MCDB	43.8	43.5	44.7	52.4	60.2	63.1	68.1	66.5	61.2	54.4	48.7	46.0	(34)	(35)
(35) Mean Daily Temperature Range	5% DB	MDBR	6.8	7.8	9.4	11.8	13.0	12.2	11.6	11.4	10.5	9.2	7.5	6.9	(35)	(36)
		MCDBR	7.4	7.9	11.8	18.7	20.3	19.0	18.7	16.5	15.1	11.8	8.3	7.5	(36)	(37)
	5% WB	MCWBR	6.4	6.4	7.1	9.3	9.1	8.4	7.7	6.9	7.1	6.9	6.5	6.6	(37)	(38)
		MCWBR	6.9	7.0	9.1	15.3	18.6	16.0	16.1	13.9	11.4	9.4	7.7	7.1	(38)	(39)
(39) Clear-Sky Solar Irradiance	taub		0.285	0.316	0.340	0.370	0.360	0.356	0.371	0.367	0.350	0.328	0.293	0.392	(39)	(40)
		taud	2.143	2.265	2.318	2.314	2.423	2.469	2.451	2.459	2.495	2.437	2.232	2.612	(40)	(41)
	Ebn at Noon		145	213	248	262	275	278	270	262	246	211	143	102	(41)	(42)
		Edh at Noon	13	21	28	34	32	31	31	29	24	18	12	9	(42)	(43)
(43) All-Sky Solar Radiation	RadAvg		70	237	574	1086	1563	1664	1476	1132	692	335	105	44	(43)	(44)
	RadStd		11	42	88	110	149	199	141	112	86	50	15	6	(44)	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	+0.56	N/A	N/A	N/A	N/A	-1.57	N/A	N/A	N/A	N/A
(47)	Regional (39 neighbors)	N/A	+1.22	N/A	N/A	N/A	N/A	-151	-221	N/A	N/A

Nomenclature: See separate page